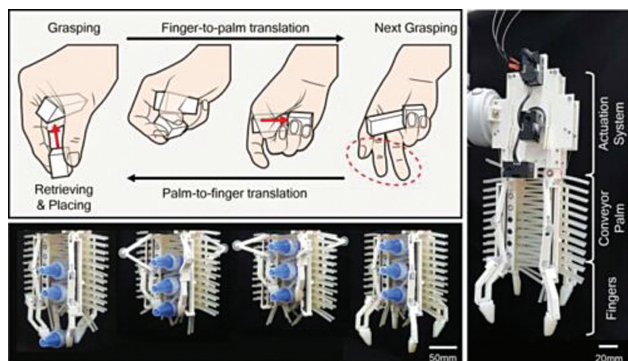




A robotic hand that can mimic human hand movements

A research team from Seoul National University has developed MOGrip, a robotic gripper inspired by human multi-object grasping techniques. MOGrip enhances pick-and-place operations by moving multiple items simultaneously and precisely placing them individually. The gripper mimics human in-hand translation, where objects



Researchers at Seoul National University proposed a gripper capable of multi-object grasping in unstructured environments (Credit: Science Robotics, 2024)

are transferred from the palm to the fingers for precise placement. MOGrip features two motor-powered fingers for grasping and a conveyor-like palm with elastic hairs for storing and transferring objects. Decoupling links simplify control by separating grasping and transferring actions. Lab tests demonstrated a 34% reduction in processing time and a 71% cut in travel distance, proving its utility in logistics, home, and unstructured tasks like bin-picking.

Fluorinated graphene for optoelectronics and energy

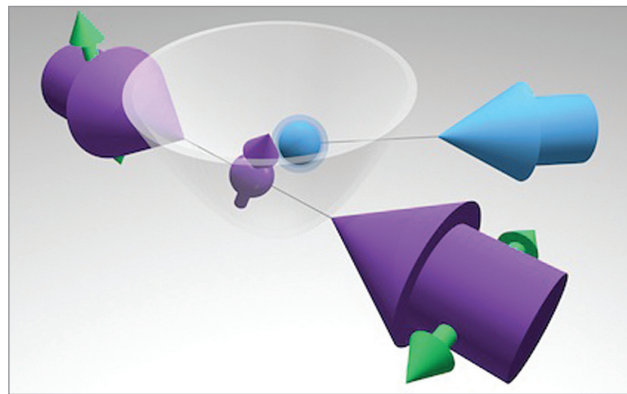
In collaboration with global experts, researchers at Tohoku University have developed an eco-friendly fluorination method to address graphene's zero-bandgap limitation. This enhances graphene for optoelectronic and energy devices without sacrificing its electrical properties. Traditional fluorination methods relied on toxic chemicals, but this new approach uses fluoropolymers to fluorinate graphene selectively under controlled conditions. The process improves photoluminescence, carrier mobility, and tunable transport properties, making it ideal for LED manufacturing, renewable energy, and flexible electronics. The team employed photoluminescence mapping and Raman spectroscopy to enhance graphene's optical properties. This innovation also integrates graphene into Van der Waals heterostructures, offering multifunctional capabilities for AI, memory storage, and smart electronics.

Faster charging for lithium-sulphur batteries

A research team at DGIST has developed a nitrogen-doped porous carbon material to enhance lithium-sulphur battery performance, addressing slow charging and stability issues. Using a magnesium-assisted thermal reduction method with ZIF-8, the team synthesised a graphitic, multiporous carbon structure that supports higher sulphur loading and improves electrolyte contact. This innovation achieved a battery capacity of 705mAh g^{-1} under rapid charging, enabling a 12-minute full charge—1.6 times faster than conventional batteries. Nitrogen doping suppressed lithium polysulfide migration, retaining 82% capacity after 1000 cycles. The design enhances sulphur utilisation and stability, advancing lithium-sulphur battery commercialisation.

Quantum simulation of molecular electron transfer

Researchers at Rice University have achieved a significant breakthrough in simulating molecular electron transfer, a crucial process in energy conversion and molecular systems. Using a trapped-ion quantum simulator, they modelled electron transfer dynamics precisely, addressing challenges like energy gaps, vibronic couplings, and envi-



A crystal with $^{171}\text{Yb}^+$ - $^{172}\text{Yb}^+$ ions is trapped in an ultra-high vacuum system. The researchers use different lasers to perform the simulation: one pair of lasers (indicated by the purple arrows) is used to simulate the coherent part of the evolution, while another laser (the blue arrow) is used to simulate and control the environment (Credit: Guido Pagano/Rice University)

ronmental dissipation. By manipulating ion crystals with laser light, they replicated a standard model of electron transfer and explored adiabatic and nonadiabatic regimes. Their findings shed light on mechanisms underlying photosynthesis and molecular devices. This research paves the way for advances in renewable energy, molecular electronics, and quantum computing, with plans to expand studies to DNA charge transport and quantum coherence.